KPI-Aware Multimodal Anomaly Detection for 5G and Open RAN Telemetry

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Abstract—Reliable anomaly detection is a deployment bottleneck for 5G/Open RAN observability and emerging self-optimizing 6G systems, where hundreds of heterogeneous KPIs, severe class imbalance, and strict alert budgets make purely numerical scores brittle. We present KAC, a deployment-oriented KPI-aware multimodal anomaly-detection framework that turns compact operator-style summaries into a learnable feedback signal over full-KPI temporal evidence. This gives the detector broad numerical coverage like modern time-series models, while retaining the sparse KPI-specific reasoning that makes operator feedback sustainable in a future operator-facing workflow. Across public telecom telemetry, enterprise-scale Open RAN observability data, and real 5G RAN production packet traces from Nokia's KPI construction pipeline, KAC consistently improves anomaly-detection performance over residual-only, deep MTSAD, and zero-shot LLM baselines. At a fixed 2.5% false-positive-rate operating regime, it substantially improves recall while remaining stable under severe class imbalance. KAC is integrated in pre-production shadow mode with Nokia's GKE-based KPI construction pipeline, where it continuously scores telemetry windows alongside an internal packet-level anomaly detector for operational evaluation.

Index Terms—telecom anomaly detection, 5G/Open RAN telemetry, multivariate time series, multimodal learning, time-series foundation models

I. Introduction

Modern cellular observability pipelines collect heterogeneous telemetry from radio access networks (RANs), cloud/platform infrastructure, and packet-level measurements. In 5G and Open RAN, where RAN functions are increasingly disaggregated, virtualized, and connected through open interfaces, these telemetry streams include fine-grained key performance indicators (KPIs) such as radio quality, throughput, latency, resource usage, and packet-flow statistics [1], [2]. These streams feed AIOps workflows for alerting, incident management, and diagnosis, yet noisy alert streams remain a persistent operational pain point [3]–[5]. The challenge is amplified in Open RAN because anomaly detectors must jointly reason over radio-layer behavior and platform-level telemetry from disaggregated network functions [1]. Recent telecom observability benchmarks further show that such data are heterogeneous, high-dimensional, and useful beyond forecasting, including anomaly detection, root-cause analysis, and multimodal reasoning [2]. We target producing reliable anomaly scores from noisy, high-dimensional telecom telemetry windows under strict false-alarm constraints.

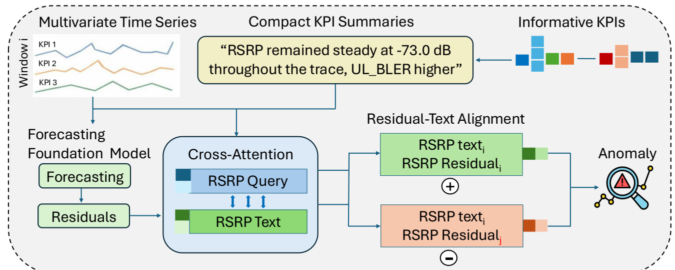

Fig. 1. Overview of KAC. "Residuals" denote the per-KPI differences between observed values and a foundation model's quantile forecasts

Multivariate time-series anomaly detection (MTSAD) has been studied extensively [6], [7], with deep models learning numerical representations of temporal behavior and inter-variable structure [8]–[13]. However, these methods typically treat variables as numerical channels, graph nodes, or attention patterns rather than as named KPIs with operational meaning. In telecom observability, KPI identity and operating context often matter as much as the magnitude of a deviation. The same numerical change may be benign for one KPI yet indicate service degradation for another, and its interpretation can depend on the surrounding operating regime [2].

Time-series foundation models (TSFMs), including Chronos/Chronos-2 [14], [15], MOMENT [16], Moirai [17], and TimesFM [18], provide strong pretrained temporal models and naturally suggest residual- or embedding-based detectors. Yet recent evidence shows that shallow forecasting-error or reconstruction-error reuse is often insufficient for anomaly detection [19]. In telecom settings, residual-only TSFM detectors are especially fragile in two cases. Context extrapolation: if an anomalous regime already begins inside the forecasting context, a strong forecaster may extrapolate it and produce small residuals even for an abnormal window. Operating-context erasure: residualization can suppress absolute KPI level; for example, a stable reference signal received power (RSRP) series may be forecast accurately both near the tower and at the cell edge, even though its absolute level is essential for interpreting block error rate (BLER), throughput, and packet-level indicators. These failure modes motivate combining full-KPI residual evidence with compact KPI-aware context rather than relying on forecast error alone.

We present KAC, a multimodal telecom anomaly detector that fuses two complementary modalities of each KPI window: a numerical modality and a textual modality. The numerical branch keeps full-KPI residual and uncertainty features from a frozen forecasting time-series foundation model, preserving broad temporal coverage across all KPIs. The textual branch encodes a compact operator-style summary that names the salient KPIs and their operating context under a token budget, reintroducing the KPI identity and absolute-level cues that residualization tends to erase. To make the two modalities reinforce each other rather than be concatenated as independent feature blocks, KAC uses contrastive learning: learnable KPI queries extract KPI-specific text representations, a shared residual projector produces matched KPI-level residual representations, and a KPI-level contrastive objective aligns the text and residual views of each KPI in a shared embedding space, so that the two modalities of a given KPI window retrieve each other within a training batch. This alignment lets sparse, KPI-specific language calibrate the dense numerical evidence instead of acting as a standalone classifier. Optional uncertainty weighting down-weights residual evidence with wide Chronos-2 prediction intervals. At inference time, KAC outputs an anomaly score from cached residual and text features; the language model is not used as the detector.

This work is grounded in both reproducible public evaluation and industrial integration. We evaluate on public 5G telemetry, enterprise-scale Open RAN telemetry, and ProdTrace-SA, a Nokia collaboration benchmark built from real 5G RAN production packet captures with eight realism-hardened injected anomaly families. KAC is integrated with Nokia's GKE-based KPI-construction pipeline in pre-production shadow mode, where it scores cached windows for offline evaluation. Code, processed public-benchmark artifacts, and evaluation scripts are released for replication.[1] Our main contributions are:

- We introduce KAC, a telecom-oriented anomaly detector that combines full-KPI residual and uncertainty features from a frozen forecasting foundation model with compact operator-style context, addressing context extrapolation and operating-context erasure in residual-only detectors.
- We propose KPI-level residual–text alignment and uncertainty-aware contrastive learning for combining temporal residual evidence with compact KPI context.
- We develop a practical text pipeline for telecom observability, including leakage-aware summary construction, label-indicative masking, train-split-only saliency, and empirical leakage checks.
- We provide a comprehensive evaluation across public 5G telemetry, enterprise-scale Open RAN telemetry, and a Nokia collaboration benchmark built from real 5G RAN production packet captures, together with low-FPR alert-budget analysis and pre-production shadow-integration evidence from Nokia's KPI-construction pipeline.

[1]Code and reproducibility artifact: https://github.com/ChimanSalavati/kac-telecom-anomaly-detection.

II. Related Work

Telecom anomaly detection has moved from manual thresholding toward AI/ML-based monitoring for 5G and O-RAN environments [20]–[22]. Recent systems emphasize reasoning across multiple telemetry layers; for example, SpotLight combines anomaly detection and localization over fine-grained RAN and platform metrics [1]. In parallel, AIOps studies identify alert and incident management as a central failure-management task, with dynamic alert suppression highlighting the persistence of noisy alert streams in operational settings [3]–[5]. These works motivate practical KPI anomaly detection, but they primarily operate on numerical telemetry and do not explicitly model KPI semantics or operator context.

Time-series foundation models such as Chronos/Chronos-2 [14], [15], MOMENT [16], Moirai [17], and TimesFM [18] provide pretrained forecasting and representation backbones for downstream time-series tasks. A common anomaly-detection strategy is to score reconstruction or forecasting errors, or to train shallow probes on frozen representations; however, recent evidence suggests that such direct reuse can be weak for anomaly detection [19]. Prompting-based LLM detectors, including SigLLM [23] and recent systematic studies of LLMs for time-series anomaly detection [24], show promise on simple cases but remain less reliable than specialized detectors in subtle regimes, motivating the use of language as contextual supervision rather than as the detector itself.

Contrastive learning is widely used to improve MTSAD representations. CAROTS constructs causality-preserving and causality-disturbing augmentations [25]; MPFormer uses multi-patch Transformer representations with contrastive learning [26]; TFCL combines time- and frequency-domain views with contrastive objectives and gated fusion [27]; and DCdetector replaces reconstruction with a dual-attention contrastive criterion between in-patch and patch-wise views [28]. More broadly, supervised contrastive learning structures embeddings using label-aware positives and negatives [29], while CLIP demonstrates the effectiveness of aligning heterogeneous modalities in a shared representation space [30]. Unlike prior MTSAD methods that contrast multiple numerical views of the same signal, KAC aligns KPI-level temporal and textual representations.

Multimodal time-series methods show that textual or contextual supervision can improve modeling beyond purely numerical representations, through time–text semantic alignment for anomaly detection [31], contrastive visual/textual views for forecasting [32], alignment of time-series embeddings with natural-language descriptions for understanding and zero-shot classification [33], LLM-generated descriptions for trajectory forecasting [34], and natural-language supervision for retrieval and channel-level grounding [35]. These methods are not designed for telecom KPI anomaly detection over full-KPI residual representations. KAC fills this gap by combining foundation-model residual and uncertainty features, compact KPI-window summaries, and KPI-level residual–text alignment in a telecom anomaly detector.

III. Method

KAC addresses the context-extrapolation and operating-context-erasure failures of residual-only detection by feeding full-KPI residuals to the numerical branch and a fixed-budget salient-KPI summary to the text branch; KPI queries and a shared residual projector align the two views, with optional interval-width uncertainty weighting. Figure 1 provides an overview of the full pipeline.

A. Problem Statement

A KPI window $\mathbf{X} \in \mathbb{R}^{T \times K}$ contains T timesteps and K KPI channels, with entry $x_{t,k}$ (or $x_{i,t,k}$ for sample i) denoting the value of KPI k at timestep t. Each window is paired with a natural-language summary s that describes salient KPI behavior or operating context. The summary is not assumed to enumerate all K KPIs: in high-dimensional telemetry, it may mention only a subset of informative indicators under a fixed token budget, while the numerical branch always receives the full multivariate window.

KAC learns a detector

$$f_\theta(\mathbf{X}, s) = \hat{p} \in [0, 1], \tag{1}$$

where $\hat{p}$ is the predicted anomaly probability.

KAC decomposes as

$$f_\theta(\mathbf{X}_i, s_i) = h_\theta(\Phi_C(\mathbf{X}_i), s_i)\,, \tag{2}$$

where Φ_C is a frozen forecasting foundation-model residual extractor with context length C, and h_θ is the trainable multimodal detector. In all reported experiments, Φ_C is instantiated with Chronos-2. The first C timesteps serve as forecasting context, so the residual (forecast-target) sequence length is $T_r = T - C$.

Throughout the method section, K denotes the number of KPI channels, d the hidden dimension, p the contrastive projection dimension, B the batch size, and L the maximum text length after tokenization.

B. Foundation-Model Residual Feature Extraction

KAC first converts each KPI window into forecast-residual features using a frozen forecasting time-series foundation model. We define Φ_C as a residual-extraction interface with context length C: given a KPI history, Φ_C returns point forecasts and, when available, prediction intervals for the target timesteps. This makes KAC agnostic to the specific forecasting backbone as long as the backbone provides the residual/uncertainty interface required by the numerical branch. In all reported experiments, we instantiate Φ_C with Chronos-2 because it provides quantile forecasts that support both residual construction and uncertainty weighting.

For sample i, KPI k, and residual timestep $\tau \in \{1, \ldots, T_r\}$, the forecasting backbone produces quantile forecasts $\hat{q}^{(0.05)}_{i,\tau,k}$, $\hat{q}^{(0.50)}_{i,\tau,k}$, and $\hat{q}^{(0.95)}_{i,\tau,k}$ for target value $x^\star_{i,\tau,k} = x_{i,\tau+C,k}$. Suppressing the common indices (i, τ, k), let $x^\star = x^\star_{i,\tau,k}$ and $\hat{q}^{(\cdot)} = \hat{q}^{(\cdot)}_{i,\tau,k}$ for each quantile level $(\cdot) \in \{0.05, 0.50, 0.95\}$. Using the same index-suppressed notation, we compute the residual feature vector

$$\boldsymbol{\rho} = [r,\ m^-,\ m^+,\ w,\ \tilde{r}], \tag{3}$$

where

$$\begin{aligned} r &= x^\star - \hat{q}^{(0.50)}, \\ m^- &= x^\star - \hat{q}^{(0.05)}, \\ m^+ &= \hat{q}^{(0.95)} - x^\star, \\ w &= \hat{q}^{(0.95)} - \hat{q}^{(0.05)}, \\ \tilde{r} &= \frac{|r|}{|x^\star| + \epsilon}, \end{aligned} \tag{4}$$

and $\epsilon > 0$ is a small constant added throughout for numerical stability. Restoring indices, we write $\boldsymbol{\rho}_{i,\tau,k}$ and denote its a-th component by $\rho^{(a)}_{i,\tau,k}$; $\rho^{(4)}_{i,\tau,k} = w$ is the raw prediction-interval width used for uncertainty weighting.

Residual features from all KPIs are stacked into $\mathbf{R}^{\text{raw}}_i \in \mathbb{R}^{T_r \times 5K}$, where each KPI occupies a contiguous block of five channels:

$$R^{\text{raw}}_{i,\tau,5(k-1)+a} = \rho^{(a)}_{i,\tau,k}. \tag{5}$$

Before training, each residual channel is z-normalized using training-split statistics to produce the numerical input tensor $\mathbf{R}_i$. Raw prediction-interval widths are retained separately for uncertainty-weighted training.

C. Compact Text Construction

KAC uses text as compact semantic side information rather than as a replacement for numerical telemetry. Here a branch refers to an input stream of the model: the numerical branch always receives the full residual tensor $\mathbf{R}_i \in \mathbb{R}^{T_r \times 5K}$, while the text branch receives a budget-limited summary s_i that may mention only a subset of salient KPIs.

When summaries are not already provided, they are generated in two stages. First, a KPI selector is fit on the training split only and ranks channels by a saliency score. In label-free settings this score can use residual magnitude, residual variance, raw KPI variability, and operator-specified priority lists; when development labels are available, a supervised selector may be fit only on training labels. Category caps can be used so that a large family of similar counters does not consume the full text budget. Second, the top-ranked KPIs for each window are described with a fixed template or offline text generator using observed levels, trends, dispersion, and non-label timing or configuration context.

This construction handles large K by compressing only the text side: one KPI is prioritized over another by the train-split saliency score and diversity constraints, but all unmentioned KPIs remain available to the numerical branch. The selector is fixed before validation/test/deployment and summaries are cached offline.

Before tokenization, we mask a fixed vocabulary $\mathcal{V}_{\text{mask}}$ of label-indicative or diagnostic terms:

$$s_i^{\text{mask}} = \text{Mask}(s_i; \mathcal{V}_{\text{mask}}). \tag{6}$$

Generated summaries never use anomaly labels or test statistics; saliency models are fit only on the training split.

For reproducibility, the artifact stores masked public-benchmark summaries, masking vocabularies, and the scripts/templates used to construct deterministic summaries. The masking vocabulary covers 30 anomaly- or diagnostic-related terms (e.g. "anomaly", "fault", "congestion") that are replaced by a mask token before encoding. For restricted data, prompt templates and generation settings are documented for audit by authorized collaborators; because the underlying PCAP-derived windows cannot be redistributed, full public end-to-end reproduction is supported on TelecomTS and SpotLight.

D. Text Encoding and KPI-Level Representations

The masked summary s_i^{mask} is tokenized to length L and encoded with DistilBERT [36]; only the last two transformer blocks are unfrozen. We use DistilBERT for deployment-efficient text encoding. The resulting token states are projected to the KAC hidden dimension, $\mathbf{E}_i^{\text{tok}} \in \mathbb{R}^{L\times d}$, and are used for both KPI-query attention and multimodal fusion.

KAC learns K KPI query vectors, $\mathbf{Q} \in \mathbb{R}^{K\times d}$, one per KPI channel. These queries attend over $\mathbf{E}_i^{\text{tok}}$ using masked multi-head attention [37], producing KPI-level text representations $\mathbf{E}_i^{\text{text}} \in \mathbb{R}^{K\times d}$. Row $\mathbf{e}_{i,k}^{\text{text}}$ corresponds to KPI k and remains context-dependent even when the compact summary does not explicitly name that KPI.

For each KPI k, the corresponding five residual channels are mapped with a shared linear projection and mean-pooled over residual timesteps to produce $\mathbf{e}_{i,k}^{\text{res}} \in \mathbb{R}^d$. Stacking over KPIs gives $\mathbf{E}_i^{\text{res}} \in \mathbb{R}^{K\times d}$.

The text-derived and residual-derived KPI representations are projected and L2-normalized into a shared p-dimensional contrastive space: $\mathbf{z}_{i,k}^{\text{text}}, \mathbf{z}_{i,k}^{\text{res}} \in \mathbb{R}^p$. These paired embeddings are used in the uncertainty-weighted KPI contrastive loss in Section III-F.

E. Gated Multimodal Fusion and Classification

The KPI-level representations support contrastive alignment, while the final anomaly score is produced from a fused sequence containing text tokens and the full-KPI residual sequence in the shared hidden space.

We first project the residual tensor to the hidden dimension, $\mathbf{U}_i \in \mathbb{R}^{T_r\times d}$, and compute a masked-mean text representation $\bar{\mathbf{e}}_i^{\text{tok}}$ from $\mathbf{E}_i^{\text{tok}}$. A feature-wise gate

$$\mathbf{g}_i = \sigma(\text{MLP}_{\text{gate}}(\bar{\mathbf{e}}_i^{\text{tok}})) \in \mathbb{R}^d \tag{7}$$

modulates each residual timestep:

$$\tilde{\mathbf{U}}_{i,\tau} = \mathbf{U}_{i,\tau} \odot \mathbf{g}_i. \tag{8}$$

The text summary therefore conditions which latent residual dimensions are emphasized before multimodal fusion.

The projected text sequence and gated residual sequence are concatenated along the sequence dimension and refined with a residual 1D convolution, dropout, and layer normalization. A learnable classification query then attends over the fused sequence to produce the window-level representation $\mathbf{h}_i \in \mathbb{R}^d$.

Finally, a two-layer MLP maps $\mathbf{h}_i$ to anomaly logit ℓ_i and predicted probability $\hat{p}_i = \sigma(\ell_i)$. In parallel, $\mathbf{h}_i$ is projected and L2-normalized to $\mathbf{z}_i^{\text{sc}} \in \mathbb{R}^p$ for supervised contrastive learning.

F. Uncertainty-Weighted Training Objective

KAC is trained with three losses: binary cross-entropy for window-level anomaly detection, supervised contrastive learning over window embeddings, and uncertainty-weighted KPI-level contrastive learning between text-derived and residual-derived KPI representations.

Given ground-truth label $y_i \in \{0,1\}$ and anomaly logit ℓ_i, we use class-weighted binary cross-entropy

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{B}\sum_i \left[w_+ y_i \log\sigma(\ell_i) + (1-y_i)\log(1-\sigma(\ell_i))\right], \tag{9}$$

where $w_+ = N^-/N^+$ is computed from training-set class counts.

Following supervised contrastive learning [29], we apply a contrastive loss to the window-level embeddings $\mathbf{z}_i^{\text{sc}}$:

$$\mathcal{L}_{\text{SupCon}} = -\frac{1}{|\mathcal{V}|}\sum_{i\in\mathcal{V}} \frac{1}{|\mathcal{P}(i)|}\sum_{j\in\mathcal{P}(i)} \log\frac{\exp(\mathbf{z}_i^{\text{sc}}\cdot\mathbf{z}_j^{\text{sc}}/\tau_{\text{cl}})}{\sum_{m\neq i}\exp(\mathbf{z}_i^{\text{sc}}\cdot\mathbf{z}_m^{\text{sc}}/\tau_{\text{cl}})}, \tag{10}$$

where $\mathcal{P}(i)$ denotes same-class positives for anchor i, $\mathcal{V}$ is the set of anchors with at least one positive in the batch, and $\tau_{\text{cl}} = 0.07$.

For each KPI k, we form a similarity matrix

$$\mathbf{S}_k = \mathbf{Z}_k^{\text{text}}(\mathbf{Z}_k^{\text{res}})^\top/\tau_{\text{cl}}, \tag{11}$$

where $\mathbf{Z}_k^{\text{text}}, \mathbf{Z}_k^{\text{res}} \in \mathbb{R}^{B\times p}$ are the batch-stacked KPI projections. We then apply symmetric InfoNCE:

$$\ell_k^{\text{KPI}} = \frac{1}{2}\left[\text{CE}_{\text{diag}}(\mathbf{S}_k) + \text{CE}_{\text{diag}}(\mathbf{S}_k^\top)\right], \tag{12}$$

where $\text{CE}_{\text{diag}}(\cdot)$ denotes the row-wise cross-entropy that treats each row's diagonal (matched-index) entry as the correct target, so that text- and residual-derived representations of the same KPI within the batch retrieve each other.

KPI-level contrastive learning should not treat every KPI equally: wide prediction intervals from the forecasting backbone indicate less reliable residual evidence. Using the raw interval-width feature $\rho^{(4)}_{i,\tau,k}$, we define

$$u_{i,k} = \frac{1}{T_r}\sum_{\tau=1}^{T_r} |\rho^{(4)}_{i,\tau,k}|, \qquad \tilde{\omega}_{i,k} = \frac{1}{u_{i,k}+\epsilon}. \tag{13}$$

Weights are normalized within each sample and averaged across the batch:

$$\omega_k = \frac{1}{B}\sum_{i=1}^{B} \frac{\tilde{\omega}_{i,k}}{K^{-1}\sum_{j=1}^{K}\tilde{\omega}_{i,j}+\epsilon}. \tag{14}$$

The uncertainty-weighted KPI contrastive loss is

$$\mathcal{L}_{\mathrm{KPI}} = \frac{\sum_{k=1}^{K}\omega_k \ell_k^{\mathrm{KPI}}}{\sum_{k=1}^{K}\omega_k+\epsilon}. \tag{15}$$

The final training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{BCE}} + \alpha\mathcal{L}_{\mathrm{SupCon}} + \beta\mathcal{L}_{\mathrm{KPI}}, \tag{16}$$

where α and β control the contributions of the two contrastive terms.

IV. Experiments

A. Experimental Setup

We evaluate KAC on three complementary telecom anomaly-detection benchmarks that differ in dimensionality, telemetry source, and class imbalance (Table I). The evaluation compares KAC with trained residual classifiers, frozen time-series foundation-model probes, deep MTSAD baselines, classical one-class detectors, a paper-aligned Open RAN pipeline, and zero-shot LLM baselines. LLMs are reported separately because they receive no task-specific training or threshold tuning. All dataset splits used in our experiments are released in our public GitHub artifact for reproducibility, with the sole exception of ProdTrace-SA, which is proprietary to Nokia and cannot be redistributed.

1) Datasets:

a) TelecomTS [2]: TelecomTS is a public 5G observability benchmark with 16 KPIs sampled at 100 ms. It contains mixed real jamming and synthetic telecom-rooted faults. We evaluate it under two scenarios, reported as Scenario 1 and Scenario 2 in Table II: Scenario 1 uses the benchmark's balanced split, while Scenario 2 uses a severely imbalanced stress split with a natural anomaly rate of approximately 3.9%.

TABLE I
Dataset characteristics after KAC preprocessing. Raw scale reports the source telemetry before windowing.

Dataset	K	T	C	T_r	Raw scale
TelecomTS	16	128	20	108	1.02M norm. / 120k anom. obs
SpotLight	452	64	20	44	100M+ telemetry points
ProdTrace-SA	64	32	8	24	116 PCAPs, 22.6M packets

b) SpotLight [1]: SpotLight is a public enterprise Open RAN telemetry benchmark with 452 KPI channels and more than 100M telemetry points at 100 ms resolution. It includes four anomaly families and six pairwise mixes, and its evaluation split is approximately class-balanced. For compact summaries, we rank KPIs with a train-only random-forest importance model on aggregated residual features from the chosen forecasting backbone, aggregate importances to KPI level with category-diversity constraints, and use deterministic templates to describe the selected KPI levels, trends, and dispersion.

c) ProdTrace-SA: ProdTrace-SA is a Nokia collaboration benchmark built from real 5G RAN production packet captures from a major telecom equipment vendor, collected from operational network environments. Because naturally labeled attack incidents are unavailable, eight realism-hardened anomaly families are injected into the production traffic: DataExfiltration, SignalingFlood, PortScan, SMPPSpam, TCPAnomalies, ExactPacketReplay, SeqNumberManipulation, and DuplicateBurst. The injected anomalies preserve production-traffic characteristics through realism-hardening procedures such as RTT sampling, inter-arrival variation, and header/payload diversity. The same Nokia GKE-based KPI-construction infrastructure converts packet traces into 64 KPI windows. For compact summaries, a label-free selector ranks packet-derived KPI channels from training windows using raw-window variability statistics, and the selected KPI statistics are summarized offline with an instruction-tuned LLM. Train, validation, and test windows are split by disjoint PCAPs to avoid capture-level leakage. ProdTrace-SA is Nokia-internal; public artifacts release the full evaluation pipeline and reproductions for public benchmarks. We use this production-traffic benchmark to evaluate sparse-positive behavior under a severe class-imbalance regime (anomaly rate $\approx 15.7\%$) and to support the pre-production shadow-integration study in Section V-A.

2) Windowing and Labels: TelecomTS provides fixed windows. For SpotLight and ProdTrace-SA, point- or record-level labels are converted to window labels by marking a window anomalous if any constituent timestep or record is anomalous. SpotLight uses non-overlapping windows with $T = 64$. ProdTrace-SA uses $T = 32$ with stride 4, resetting window construction at PCAP boundaries.

3) Evaluation Protocol: We report threshold-free ranking metrics (AUROC and AP) and thresholded operating-point metrics (Precision, Recall, and F1). To ensure

protocol fairness, all trainable methods use the same train/validation/test discipline. All trainable methods are fit on the training split only, with validation used exclusively for early stopping, model selection, and threshold calibration. Unsupervised methods similarly use validation only for threshold selection. Results are reported on the held-out test split across five seeds.

For thresholded Precision, Recall, and F1, we select a single decision threshold for each method and seed on the validation split by maximizing validation F1, and then apply that threshold unchanged to the test split. For operational alert-budget analysis, we additionally calibrate fixed-FPR thresholds on validation normal windows and evaluate them on the test split. Unless otherwise stated, fixed-budget results use a 2.5% validation FPR target, corresponding to approximately 25 false alerts per 1,000 normal windows. Results are reported as mean±std across seeds; for ProdTrace-SA we additionally report bootstrap confidence intervals because of its small positive test count. In all result tables, bold marks the best value and underline the second-best value within each metric column.

4) Baselines: We compare against five baseline families. Residual and foundation-model baselines include Chronos-2+LR, QR-TAN, and frozen foundation-model embeddings from MOMENT [16], TOTO [38], and Mantis [39] with a trained linear classification head (+linear). Chronos-2+LR trains logistic regression on aggregated Chronos-2 residual features, while QR-TAN (ctx=20) is a residual-only temporal attention network using the same forecasting context as KAC.

End-to-end supervised baselines include PatchTST-classification [40], iTransformer [41], GPT4TS/OFA [42], ModernTCN-classification [43], and UniTS [44]. Deep MTSAD and one-class baselines include DCdetector [28], TimesNet [45], ModernTCN [43], MEMTO [46], D3R [47], Isolation Forest [48], and LOF [49]. We also evaluate a simple relative-error rule that flags a window when the fraction of KPI-timestep cells with Chronos-2 relative forecast error $\tilde{r} = |r|/(|x^\star| + \epsilon)$ above 0.3 exceeds a validation-tuned threshold. Finally, we include the SpotLight JVGAN–MRPI pipeline [1] and zero-shot LLM baselines that receive KPI names and values without task-specific training or calibration.

5) Implementation Details: KAC uses hidden dimension $d = 64$, contrastive dimension $p = 128$, four attention heads, dropout 0.1, maximum token length $L = 128$, and contrastive temperature $\tau_{\mathrm{cl}} = 0.07$. For the Chronos-2 instantiation used in all reported experiments, residuals and text summaries are cached before detector training. We train with AdamW [50], batch size 16, weight decay 10^{-2}, gradient clipping at 1.0, and early stopping on validation F1. The unfrozen DistilBERT blocks use learning rate 2×10^{-5}; all remaining KAC parameters use 5×10^{-4} on public benchmarks and 10^{-3} on ProdTrace-SA. For public benchmarks we use $(\alpha, \beta) = (1, 1)$; for ProdTrace-SA we use $(0.1, 0.3)$.

Code, preprocessing scripts, and evaluation configuration are released with the artifact.

B. Results on TelecomTS

Table II shows that on the balanced TelecomTS split KAC attains the highest Precision (0.961), F1 (0.960), AUROC (0.988), and AP (0.988) among all supervised and unsupervised methods. The largest gains over residual-only and foundation-model baselines come from the compact text branch: in this relatively low-dimensional 16-KPI setting, operator-style summaries provide operating-context information that is difficult to recover from residuals alone.

Under the severely imbalanced stress split, KAC achieves the highest Precision (0.900) and ties for the highest F1 (0.800), while Mantis+linear attains the highest AUROC (0.995) and AP (0.926). This suggests that frozen foundation-model embeddings remain competitive for ranking under extremely small anomaly priors, whereas KAC retains the strongest threshold-based detection performance. Overall, the results support the central design motivation of KAC: combining full-KPI residual evidence with compact operating-context signals improves threshold-based anomaly detection beyond residual-only TSFM detectors and strong end-to-end supervised baselines.

Table III evaluates detectors at a fixed low-FPR operating point on TelecomTS corresponding to approximately 25 false alerts per 1,000 normal windows. KAC achieves the highest recall at this alert budget while maintaining comparable precision. This operating regime is important in practice because telecom operators often tune anomaly detectors by alert volume rather than by full-curve averages.

We report fixed-budget analysis only on TelecomTS. SpotLight is class-balanced, so threshold-based F1 already reflects a meaningful operating point and AUROC/AP characterize performance across the full operating range. ProdTrace-SA's test split is only $\sim$15.7% positive, making fixed-FPR recall estimates too unstable for reliable method ranking. We therefore rely on AUROC/AP for ProdTrace-SA.

Table IV reports zero-shot frontier LLMs operating directly on raw KPI matrices and KPI names. This is not intended as a fairness-matched comparison to a trained detector; rather, it evaluates the alternative design of delegating anomaly detection to an LLM without task-specific training, threshold calibration, or cached residual features. Exact prompts, model identifiers, evaluation dates, decoding parameters, parsing rules, and cached responses are released in the artifact for reproducibility. Across all benchmarks, zero-shot prompting remains substantially weaker than trained telecom-specific detectors, particularly under subtle or highly imbalanced regimes. These results support the KAC design choice of using

TABLE II
Experimental results on TelecomTS under balanced and imbalanced test settings.

	Balanced test set					Imbalanced test set				
Method	Prec.	Rec.	F1	AUROC	AP	Prec.	Rec.	F1	AUROC	AP
Supervised methods										
QR-TAN (ctx=20)	.954 ± .025	.894 ± .018	.923 ± .007	.974 ± .007	.978 ± .005	.738 ± .092	.700 ± .000	.716 ± .042	.919 ± .026	.762 ± .035
Chronos-2 + LR	.916 ± .000	.887 ± .000	.901 ± .000	.967 ± .000	.970 ± .000	.643 ± .000	.900 ± .000	.750 ± .000	.913 ± .000	.842 ± .000
MOMENT + linear	.806 ± .026	.775 ± .034	.789 ± .013	.895 ± .009	.901 ± .010	.132 ± .022	.620 ± .130	.218 ± .038	.796 ± .016	.245 ± .027
Mantis + linear	.931 ± .000	.927 ± .000	.929 ± .000	.980 ± .000	.982 ± .000	.667 ± .000	1.00 ± .000	.800 ± .000	.995 ± .000	.926 ± .000
TOTO + linear	.715 ± .019	.807 ± .042	.758 ± .010	.840 ± .003	.854 ± .003	.250 ± .000	.200 ± .000	.222 ± .000	.591 ± .021	.224 ± .029
PatchTST	.872 ± .020	.846 ± .017	.859 ± .009	.930 ± .005	.934 ± .007	.204 ± .027	.380 ± .098	.264 ± .045	.770 ± .053	.319 ± .086
iTransformer	.922 ± .015	.874 ± .016	.897 ± .008	.963 ± .003	.965 ± .004	.447 ± .044	.520 ± .040	.478 ± .020	.861 ± .022	.618 ± .017
GPT4TS	.837 ± .027	.158 ± .026	.264 ± .037	.780 ± .015	.769 ± .012	.131 ± .120	.580 ± .214	.172 ± .108	.606 ± .069	.204 ± .077
ModernTCN-cls	.872 ± .041	.842 ± .049	.855 ± .009	.945 ± .003	.950 ± .003	.595 ± .094	.360 ± .080	.435 ± .033	.897 ± .023	.574 ± .026
UniTS	.946 ± .027	.911 ± .018	.928 ± .008	.974 ± .005	.978 ± .005	.537 ± .087	.580 ± .040	.555 ± .060	.766 ± .080	.528 ± .081
KAC (Ours)	.961 ± .018	.960 ± .010	.960 ± .004	.988 ± .002	.988 ± .004	.900 ± .056	.720 ± .045	.800 ± .050	.934 ± .038	.839 ± .035
Unsupervised and one-class methods										
SpotLight	.508 ± .008	.973 ± .012	.667 ± .004	.581 ± .022	.560 ± .015	.167 ± .119	.500 ± .255	.216 ± .110	.715 ± .100	.201 ± .094
D3R	.599 ± .060	.862 ± .122	.700 ± .016	.668 ± .033	.641 ± .018	.367 ± .182	.500 ± .071	.395 ± .101	.827 ± .030	.391 ± .038
Relative-error rule	.518 ± .000	.976 ± .000	.677 ± .000	.651 ± .000	.651 ± .000	.200 ± .000	.100 ± .000	.133 ± .000	.623 ± .000	.199 ± .000
Isolation Forest	.608 ± .085	.775 ± .125	.670 ± .014	.677 ± .021	.710 ± .019	.227 ± .030	.660 ± .055	.337 ± .038	.886 ± .014	.512 ± .042
MEMTO	.731 ± .034	.608 ± .063	.662 ± .035	.721 ± .017	.729 ± .019	.274 ± .112	.720 ± .084	.382 ± .112	.811 ± .028	.440 ± .020
TimesNet	.508 ± .004	.923 ± .021	.655 ± .005	.625 ± .005	.592 ± .005	.109 ± .034	.200 ± .122	.130 ± .035	.719 ± .016	.147 ± .035
ModernTCN	.506 ± .002	.915 ± .010	.652 ± .004	.627 ± .006	.592 ± .007	.090 ± .008	.380 ± .045	.146 ± .014	.714 ± .004	.131 ± .007
DCdetector	.517 ± .013	.816 ± .095	.631 ± .025	.537 ± .015	.536 ± .016	.017 ± .024	.180 ± .249	.031 ± .043	.456 ± .029	.047 ± .002
LOF	.545 ± .000	.644 ± .000	.590 ± .000	.607 ± .000	.624 ± .000	.053 ± .000	.400 ± .000	.094 ± .000	.600 ± .000	.068 ± .000

TABLE III
Fixed alert-budget evaluation on TelecomTS. All methods are evaluated at 2.5% FPR, corresponding to 25 false alerts per 1,000 normal windows.

Method	FPR budget	Recall	Prec.	False alerts /1k normal
QR-TAN (ctx=20)	2.5%	.85	.971	25
Chronos-2 + LR	2.5%	.80	.970	25
KAC (Ours)	2.5%	.93	.974	25

TABLE IV
Zero-shot LLM prompting versus trained KAC.

Dataset	Model	F1	AUROC	AP
TelecomTS	Claude Opus 4.7	.384	.689	.613
	GPT-5	.607	.651	.612
	Gemini 2.5 Pro	.453	.523	.543
	KAC (Ours)	.960	.988	.988
ProdTrace-SA	Claude Opus 4.7	.250	.627	.521
	GPT-5	.273	.377	.127
	Gemini 2.5 Pro	.293	.690	.346
	KAC (Ours)	.742	.933	.876
SpotLight	Claude Opus 4.7	.567	.619	.532
	GPT-5	.641	.692	.616
	Gemini 2.5 Pro	.602	.660	.639
	KAC (Ours)	.950	.986	.984

language as a compact contextual side channel rather than relying on end-to-end LLM reasoning for anomaly detection.

SpotLight stresses high-dimensional Open RAN telemetry and strong residual structure [1]. Table V shows that KAC achieves the strongest F1 and Recall among all evaluated methods, leading the next strongest baseline (iTransformer) by 0.010 F1.

The high-dimensional regime highlights a different behavior than TelecomTS. End-to-end-trained patch, channel-token, and convolutional classifiers remain highly competitive, but KAC's combination of full-KPI residual evidence and KPI-level multimodal alignment retains the strongest threshold-based detection performance. Unlike TelecomTS, the primary KAC gain on SpotLight comes from KPI-level contrastive alignment rather than from the text content itself: the text branch acts mainly as a training-time alignment signal, while the full residual tensor carries the dominant detection evidence at inference time.

TABLE V
Experimental results on SpotLight.

Method	Prec.	Rec.	F1	AUROC	AP
Supervised					
QR-TAN (ctx=20)	.927 ± .005	.945 ± .024	.936 ± .011	.978 ± .009	.978 ± .007
Chronos-2 + LR	.944 ± .000	.924 ± .000	.934 ± .000	.978 ± .000	.982 ± .000
MOMENT + linear	.922 ± .014	.847 ± .049	.882 ± .021	.965 ± .003	.957 ± .005
Mantis + linear	.923 ± .000	.945 ± .000	.934 ± .000	.967 ± .000	.973 ± .000
TOTO + linear	.831 ± .012	.812 ± .008	.821 ± .004	.913 ± .001	.915 ± .001
PatchTST	.934 ± .008	.944 ± .030	.939 ± .015	.987 ± .004	.987 ± .003
iTransformer	.926 ± .010	.954 ± .017	.940 ± .004	.979 ± .002	.969 ± .005
GPT4TS	.845 ± .039	.657 ± .047	.737 ± .027	.871 ± .018	.856 ± .025
ModernTCN-cls	.945 ± .003	.900 ± .024	.921 ± .011	.983 ± .003	.983 ± .002
UniTS	.924 ± .015	.918 ± .029	.920 ± .009	.979 ± .002	.977 ± .003
KAC (Ours)	.939 ± .006	.961 ± .012	.950 ± .004	.986 ± .006	.984 ± .006
Unsupervised / one-class					
SpotLight pipeline	.658 ± .010	.910 ± .018	.763 ± .009	.773 ± .017	.720 ± .037
ModernTCN	.751 ± .010	.873 ± .050	.807 ± .027	.877 ± .018	.871 ± .015
MEMTO	.683 ± .003	.933 ± .026	.788 ± .010	.784 ± .013	.707 ± .041
D3R	.679 ± .002	.938 ± .019	.788 ± .008	.777 ± .002	.692 ± .008
Relative-error rule	.714 ± .000	.866 ± .000	.783 ± .000	.779 ± .000	.708 ± .000
TimesNet	.741 ± .015	.820 ± .007	.779 ± .006	.858 ± .011	.849 ± .019
Isolation Forest	.682 ± .007	.823 ± .062	.745 ± .023	.813 ± .010	.775 ± .019
DCdetector	.726 ± .061	.671 ± .230	.673 ± .091	.804 ± .005	.758 ± .011
LOF	.500 ± .000	.024 ± .000	.046 ± .000	.815 ± .000	.731 ± .000

C. Results on ProdTrace-SA

ProdTrace-SA evaluates detector behavior under severe class imbalance using its packet-derived KPI traces processed through Nokia's KPI construction pipeline.

Table VI shows that KAC achieves the strongest mean F1 (0.742), AP (0.876), and tied-best AUROC (0.933) among the evaluated methods. These results indicate that KPI-level multimodal alignment remains effective under sparse-positive conditions representative of pre-production telecom monitoring. Although confidence intervals remain wide because of the sparse-positive regime, KAC avoids the collapse and threshold-transfer failures observed in several supervised and one-class baselines while maintaining stable threshold-based performance.

TABLE VI
Experimental results on ProdTrace-SA.

Method	F1	AUROC	AP
Supervised			
QR-TAN (ctx=20)	.737 $\pm$.078	.905 $\pm$.056	.815 $\pm$.085
Chronos-2 + LR	.500 $\pm$.000	.637 $\pm$.000	.518 $\pm$.000
Residual+Level (LR)	.378 $\pm$.000	.669 $\pm$.000	.469 $\pm$.000
TOTO + linear	.047 $\pm$.079	.409 $\pm$.064	.135 $\pm$.014
PatchTST	.502 $\pm$.080	.876 $\pm$.064	.821 $\pm$.094
iTransformer	.411 $\pm$.081	.933 $\pm$.016	.870 $\pm$.018
GPT4TS	.214 $\pm$.141	.451 $\pm$.146	.387 $\pm$.138
KAC (Ours)	.742 $\pm$.077	.933 $\pm$.048	.876 $\pm$.066
Unsupervised / one-class			
Relative-error rule	.615 $\pm$.000	.797 $\pm$.000	.572 $\pm$.000
MEMTO	.540 $\pm$.097	.918 $\pm$.003	.708 $\pm$.056
TimesNet	.479 $\pm$.064	.851 $\pm$.056	.580 $\pm$.049
ModernTCN	.459 $\pm$.030	.914 $\pm$.005	.700 $\pm$.027
D3R	.455 $\pm$.076	.823 $\pm$.131	.666 $\pm$.074
LOF	.438 $\pm$.000	.875 $\pm$.000	.683 $\pm$.000
Isolation Forest	.276 $\pm$.030	.593 $\pm$.122	.293 $\pm$.126
DCdetector	.222 $\pm$.037	.490 $\pm$.080	.176 $\pm$.046
SpotLight pipeline [1]	.265 $\pm$.026	.511 $\pm$.049	.207 $\pm$.055

D. Paired Significance

To avoid over-interpreting small F1 gaps under sparse positives, we report paired KAC−baseline ΔF1 with 95% bootstrap confidence intervals over five seeds (Table VII); positive values favor KAC, and a CI that excludes zero indicates a seed-consistent advantage. We do not run formal significance tests because five paired seeds are underpowered. KAC's threshold-F1 lead is consistent and CI-separated on TelecomTS against all four baselines, and is large and CI-separated on ProdTrace-SA against iTransformer, PatchTST, and Mantis.

The only case where the CI includes zero is KAC versus QR-TAN on ProdTrace-SA. Although KAC remains ahead in mean F1 (+0.028), the bootstrap CI [−0.080, 0.114] crosses zero, indicating that the observed difference is not reliably separable from seed-level variation in this sparse-positive setting. This result is consistent with QR-TAN already operating on the same Chronos-2 residual evidence used by KAC.

E. Component Ablation

Table VIII isolates the contribution of the compact text branch and KPI-level contrastive alignment. V1 uses residual-only gated fusion, V2 adds text fusion, and V3 corresponds to the full KAC model with KPI-level alignment. On TelecomTS, the largest gain comes from adding the text branch, indicating that operator-style summaries recover operating-context information that is weakly expressed in residuals alone. On SpotLight, the additional gain from V2 to V3 shows that KPI-level alignment becomes increasingly important in high-dimensional telemetry, where direct text conditioning is diluted across hundreds of KPI channels.

TABLE VII
Paired KAC−baseline ΔF1 (mean and 95% bootstrap CI over five seeds; positive favours KAC).

Baseline	ΔF1 TelecomTS	ΔF1 ProdTrace-SA
QR-TAN	+.034 [.024, .042]	+.028 [−.080, .114]
iTransformer	+.060 [.050, .069]	+.355 [.309, .401]
PatchTST	+.098 [.089, .105]	+.263 [.221, .305]
Mantis	+.027 [.020, .035]	+.229 [.190, .268]

a) Is it just absolute KPI levels?: A natural concern is that KAC's gains arise from reintroducing absolute KPI levels removed by residualization rather than from language alignment. To test this, we add a non-language baseline (Residual+Level) that augments Chronos residual aggregates with raw per-KPI window statistics and train-split saliency features, without a text branch. Although adding absolute levels improves over residual-only detection, it remains below KAC on all three benchmarks and falls substantially behind on ProdTrace-SA (0.378 vs. 0.742 F1; Table VI). These results indicate that KAC's gains cannot be explained solely by recovering absolute KPI levels.

Uncertainty weighting contributes less than text fusion or KPI-level alignment on the public benchmarks, but consistently improves stability under high-variance conditions. We therefore use it primarily as a robustness mechanism in the ProdTrace-SA evaluation.

F. Text Leakage and Context Recovery Analysis

TelecomTS descriptions originate from anomaly-conditioned prompts, so we evaluate whether the text branch exploits label shortcuts rather than learning meaningful KPI context. A keyword audit finds no explicit anomaly-type tokens in the descriptions. Moreover, the performance benefit of text survives aggressive masking of label-indicative patterns, while label-independent KPI-derived summaries constructed from raw KPI statistics still improve over the residual-only baseline. These results suggest that the text branch contributes information beyond explicit anomaly labels.

We further evaluate text–residual alignment through contradictory-text controls. Replacing anomaly descriptions with normal descriptions substantially reduces F1, while increasing the text budget from 128 to 512 tokens degrades performance and makes the model largely insensitive to text permutations. Together, these results indicate that the text branch therefore acts mainly as a

TABLE VIII
Component ablation on benchmarks. V1 removes text, V2 adds text fusion, and V3 adds KPI-level contrastive alignment.

Dataset	Variant	Precision	Recall	F1	AUROC	AP
TelecomTS	V1: Residual-only	0.943 $\pm$0.015	0.895 $\pm$0.016	0.918 $\pm$0.010	0.972 $\pm$0.004	0.976 $\pm$0.002
	V2: + Text fusion	0.967 $\pm$0.010	0.953 $\pm$0.011	0.960 $\pm$0.007	0.987 $\pm$0.004	0.988 $\pm$0.005
	V3: KAC (ours)	0.961 $\pm$0.016	0.960 $\pm$0.009	0.960 $\pm$0.004	0.988 $\pm$0.002	0.988 $\pm$0.003
SpotLight	V1: Residual-only	0.920 $\pm$0.008	0.953 $\pm$0.013	0.937 $\pm$0.007	0.975 $\pm$0.006	0.968 $\pm$0.009
	V2: + Text fusion	0.931 $\pm$0.012	0.951 $\pm$0.014	0.940 $\pm$0.010	0.974 $\pm$0.011	0.967 $\pm$0.022
	V3: KAC (ours)	0.939 $\pm$0.006	0.961 $\pm$0.011	0.950 $\pm$0.003	0.986 $\pm$0.005	0.984 $\pm$0.006

TABLE IX
Operational cost of the KAC pre-production shadow scorer. Left: parameter breakdown. Right: CPU execution latency on cached residual/summary inputs

Component	Params
DistilBERT, frozen	52.19 M (78.5%)
DistilBERT, last 2	14.18 M (21.3%)
KAC fusion + heads	0.14 M (0.21%)
Learnable queries	1,088 (0.00%)
Total	66.51 M
Trainable	14.32 M (21.5%)

Batch	Med. (ms)	p95 (ms)	Tput. (w/s)
1	14.7	17.7	68.0
16	156.8	158.5	102.0
64	572.4	608.0	111.8

contextual calibration signal that suppresses false positives under different operating regimes rather than a source of label leakage.

V. Pre-Production Shadow Integration and Operational Cost

KAC is designed for low-cost pre-production shadow integration in telecom KPI pipelines. Foundation-model residual extraction, instantiated with Chronos-2 in the current shadow pipeline, and summary generation are performed offline or cached; the shadow scorer does not make online LLM calls. The latency in Table IX measures the KAC scoring path on cached residual/summary inputs, including DistilBERT text encoding and lightweight fusion/classification heads, but excluding offline Chronos-2 extraction and summary generation. Measurements are reported for CPU execution, reflecting the current shadow-scoring configuration and demonstrating that the online KAC scorer does not require GPU allocation. The KAC-specific modules account for only $\sim$0.21% of parameters beyond the DistilBERT backbone. Single-window scoring is below the 100 ms telemetry cadence used in the evaluated benchmarks (batch-1 median 14.7 ms, p95 17.7 ms), while batching improves throughput at the cost of higher batch latency. In the shadow configuration, Chronos-2 extraction is parallelized across KPI streams, and text-summary refresh is event-triggered rather than per-window, amortizing language-processing cost over operational intervals.

A. Shadow Integration Status

KAC is integrated in pre-production shadow mode with Nokia's GKE-based KPI construction pipeline, which builds per-window KPI tensors and cached operator-style summaries from 5G RAN captures. The infrastructure uses Kubernetes-based telemetry services including Kong, NATS, and MinIO for ingestion, streaming, and object storage.

In the current shadow configuration, KAC runs alongside Nokia's internal packet-level anomaly detector. The pipeline passes KPI windows to KAC, cached residuals and summaries are loaded, and the resulting anomaly scores are logged for offline comparison against the internal detector. KAC does not currently promote incidents, trigger operator-facing alerts, or initiate mitigation.

The pre-production evidence reported in this paper focuses on four operational aspects: detection quality across heterogeneous telecom benchmarks; behavior under fixed low-FPR alert budgets; robustness under severe class imbalance; and operational cost, including CPU execution latency and parameter footprint (Table IX). Together, these results support deployment compatibility and stable pre-production behavior without online LLM inference. We deliberately do not claim operator-facing impact: detection delay, alert-volume reduction, incident-level precision/recall, and head-to-head comparison against Nokia's internal packet-level anomaly detector require a controlled operator-facing rollout and are left to future deployment studies (Section VI).

VI. Limitations and Future Work

KAC improves threshold-based detection and low-FPR behavior, but it does not dominate every metric. Mantis+linear remains stronger on AUROC/AP in the imbalanced TelecomTS stress split, and PatchTST remains strongest on AUROC/AP for SpotLight, showing that frozen or end-to-end numerical baselines can remain competitive on ranking metrics. ProdTrace-SA's anomaly labels are generated through realism-hardened injection of multiple anomaly families rather than naturally occurring operator-labeled incidents, and the sparse-positive regime yields wide confidence intervals. Because ProdTrace-SA and the shadow pipeline are Nokia-internal, they provide supportive industrial evidence rather than independent public production validation; TelecomTS and SpotLight provide the fully reproducible public evidence base. Finally, the Nokia integration is limited to pre-production shadow mode: operator-facing alert-volume reduction, detection delay, incident-level precision/recall, operator feedback, and head-to-head comparison with existing production monitoring systems require a rollout. Future work will evaluate KAC on naturally occurring operational incidents, broader fault scenarios, and operator-facing deployments.

VII. Conclusion

KAC is a telecom anomaly detector that aligns foundation-model residual evidence with compact, cached KPI-aware context via KPI-level residual–text alignment and uncertainty-aware contrastive learning, without online LLM inference. Across public 5G telemetry, enterprise-scale Open RAN observability data, and production-traffic-derived KPI benchmarks, it consistently improves detection in the low-FPR regimes relevant to telecom monitoring, and it is integrated in pre-production shadow mode with Nokia's GKE-based KPI-construction pipeline. These results support KPI-aware multimodal alignment as a practical, scalable direction for telecom observability.

GenAI Usage Disclosure

We distinguish three uses of generative AI in this work. (i) Manuscript preparation: generative AI was used only for grammar checking and sentence clarification of author-written text. (ii) Offline dataset text construction: for ProdTrace-SA, the compact operator-style window summaries used by KAC's text branch were generated offline with an instruction-tuned LLM. (iii) Experimental baselines: the zero-shot results use LLM APIs (Claude, GPT, Gemini) purely as comparison baselines, with no task-specific training.

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